

Executive Summary

This study explored the Tesco 1.0 dataset (Luca et al., 2020) and the Regional gross disposable household income dataset (Trevor, 2022). Then provided two insights of grocery purchases from the Tesco 1.0 dataset in terms of the difference based on the gender population and fruit and vegetable consumption, and an insight from the income dataset to show the area with the highest income, as well as an insight after combining two datasets together to show the correlation.

Tesco 1.0 dataset

The Tesco 1.0 dataset consists of monthly and a year of grocery purchase data acquired anonymously from 1.6 million Clubcard customers who made in-store purchases in 411 Tesco stores in Greater London between January 1st and December 31st, 2015. The data was then integrated with census data and divided into four levels: LSOA, MSOA, Ward, and Borough. Data has been computed and shown statistically to indicate the product consumption in each location as well as the diversity of goods purchased. However, this data has some biases and limitations as it is only from Tesco in-store purchases made by Clubcard holders and does not show the number of people who will consume this purchase, so it does not accurately reflect food and nutrient consumption as an individual or the overall population of those areas.

Scatter plots were created to represent the insights. The first insight is showing the difference in grocery purchases between areas with more males and females, selecting only the top five areas with the most male population and the top five areas with the most female population to simplify the visualisation and show the most difference. The results of this analysis indicate fascinating findings in two categories: dairy and ready meals, which may be summarised as a larger purchase of dairy from areas with a high male population, but a larger purchase of ready meals from areas with a high female population.

The second insight indicates the difference in grocery purchases between the top five areas with the largest and lowest fruit and vegetable intake. This research reveals that the intake of grains, soft drinks, and sweets has a clear inverse relationship with the consumption of fruits and vegetables.

Regional gross disposable household income (GDHI) dataset

This regional GDHI dataset contains data on London's gross income per head for various years (1997-2020), representing the London boroughs. A heat map was created to illustrate the level of income in each area. The result shows that the area with the highest income of all time is E09000001 (the city of London), and the year with the highest income in the city of London is 2015, which will be utilised in finding the correlation when combined with the Tesco 1.0 dataset.

Combining two datasets

The study 'US Adults' Fruit and Vegetable Intakes, 1989 to 1991: A Revised Baseline for the Healthy People 2000 Objective' (Susan et al., 1995) shows a correlation between fruit and vegetable intake and income. Based on the two datasets mentioned earlier, a line chart was constructed to determine whether this discovery is still valid in 2015. As a result, the line chart indicates a tiny spike in fruit and vegetable purchases in areas with high incomes and a slight decrease in those with low incomes.

To summarise, fruit and vegetable intake and income are correlated in the Tesco 1.0 dataset and the Regional gross disposable household income dataset. The relationship is consistent with the findings of the 1995 study publications.

Bibliography

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